

Trade Finance Exception and Credit Utilization Control Platform

Designing a Better Way to Detect, Investigate and Resolve Transaction Exceptions

Portfolio Case Study: This project uses entirely synthetic data, fictional system names and simulated business scenarios. It is based on my understanding of Trade Finance processes, system integrations, reconciliation, testing and production support. It does not contain confidential information, internal documentation, screenshots, data or proprietary system designs from any employer or client.

1. Executive Business Case

The Problem

A Trade Finance transaction does not always stay within one application.

A transaction may be created or updated in a Trade Finance platform and then passed to a lending or credit platform so that the corresponding facility utilization can be updated.

When everything works correctly, this happens without manual intervention.

The problem begins when it does not.

A message may arrive late.

It may fail completely.

The same message may be processed twice.

The transaction amount may be updated in one system but not reflected correctly in another.

An acknowledgement may never arrive.

Operations teams then have to work out what happened by checking multiple systems, timestamps, message records and transaction details.

This creates two problems.

The first is operational: people spend time finding and reconciling exceptions.

The second is control-related: a transaction and its credit utilization can temporarily show different values across systems.

Who Is Affected?

Trade Finance Operations

Investigates failed and mismatched transactions.

Credit Risk

Needs confidence that facility utilization is accurate.

Technology and Production Support

Investigates integration and application failures.

Product Owner

Prioritizes improvements and owns the business process.

Compliance and Controls

Needs evidence that exceptions are identified and resolved appropriately.

Management

Needs visibility into exception volumes, recurring causes and SLA performance.

Proposed Solution

I designed a centralized **Trade Finance Exception and Credit Utilization Control Platform**.

The platform would monitor transactions moving between two fictional systems:

TradeFlow – Trade Finance transaction platform

CreditCore – Lending and facility-utilization platform

An integration layer, **MessageBridge**, represents the messaging infrastructure connecting them.

The platform would:

- monitor transaction movement between systems;
- identify missing acknowledgements;
- detect duplicate messages;
- detect delayed processing;
- compare transaction and utilization amounts;
- create exception cases automatically;
- prioritize cases based on business impact;
- assign exceptions to the appropriate team;
- track SLA deadlines;
- maintain investigation history;
- provide reconciliation controls;

- produce operational and management reporting.

An optional summarization feature would help investigators understand complicated exceptions, but it would **not make financial decisions or close critical cases automatically**.

Alternatives Considered

Option 1 – Continue With Manual Reconciliation

Lowest implementation cost, but the underlying problem remains.

Operations still need to search across systems and spreadsheets.

Decision: Rejected.

Option 2 – Build Separate Monitoring Inside Each Application

This would improve visibility within individual systems but would not provide a complete transaction journey.

A transaction can appear successful in TradeFlow while failing later in MessageBridge or CreditCore.

Decision: Not selected as the primary solution.

Option 3 – Centralized Exception and Reconciliation Platform

This creates one view across the transaction lifecycle and separates exception management from individual source applications.

Decision: Selected.

The main reason was traceability.

Operations should be able to start with a Transaction ID and understand what happened across the complete flow without manually reconstructing it.

Projected Benefits

Because this is a portfolio case study rather than a production implementation, these are **targets, not achieved results**.

Measure	Proposed Target
Manual reconciliation effort	30% reduction
Exception detection	Within 5 minutes for monitored events
SLA compliance	≥95%
Reconciliation accuracy	≥99.5%
Duplicate processing	Prevent 100% of detected exact duplicates
Critical exception visibility	Within 5 minutes
STP rate	≥95%

Major Risks

The solution does not remove every operational risk.

Remaining risks include:

- incorrect upstream data;
- network or infrastructure outages;
- incorrect business-rule configuration;
- false-positive exceptions;
- reference-data inconsistencies;
- currency and timing differences;
- downstream system unavailability;
- inappropriate user access;
- incorrect automated summaries.

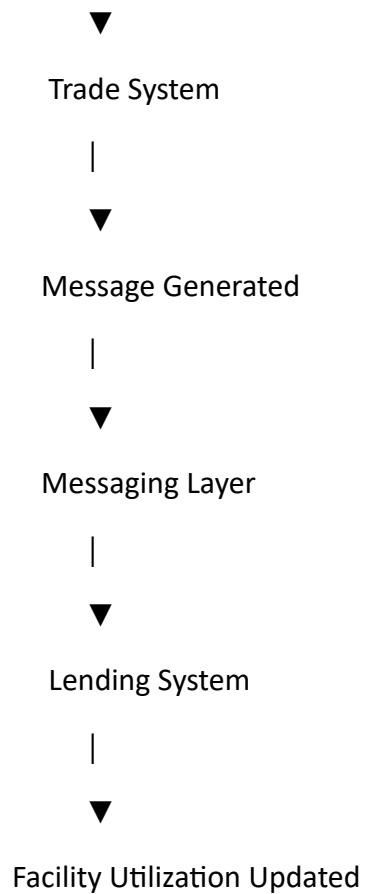
Controls are therefore required at the data, workflow and access levels.

2. AS-IS Process

Current Process

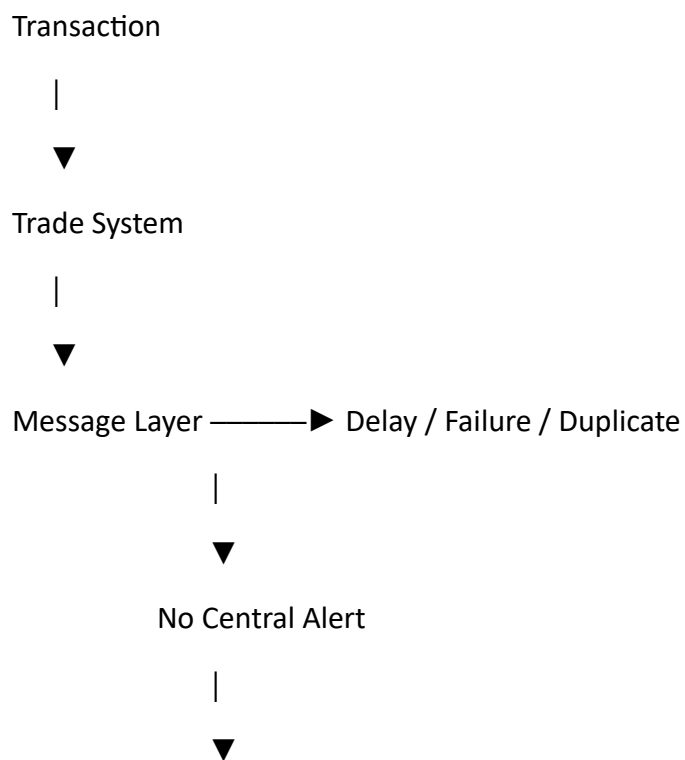
Trade Transaction Created

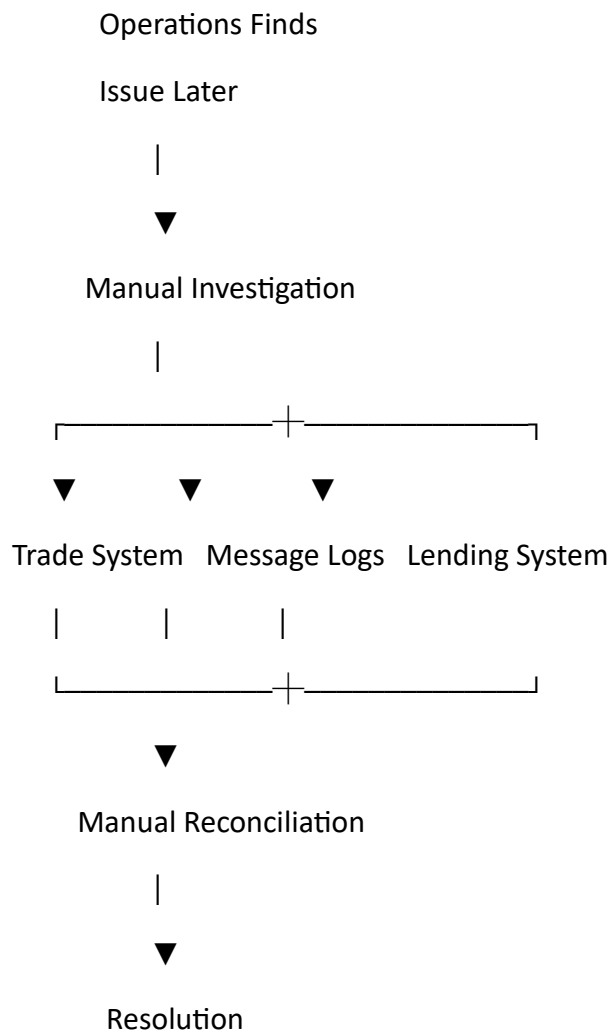
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This process works well when the transaction completes successfully.

The weakness appears when something fails.





Main AS-IS Gaps

Reactive Detection

Operations may discover problems during reconciliation instead of when the failure occurs.

Fragmented Investigation

Information sits across multiple applications and logs.

Manual Prioritization

A ₹5,000 discrepancy and a ₹50 million discrepancy should not necessarily receive the same operational priority.

Limited SLA Visibility

Teams may know when a case was created without having a reliable view of how close it is to breaching its resolution SLA.

Weak Root-Cause Reporting

Cases can be resolved individually without revealing that the same integration problem has occurred repeatedly.

3. TO-BE Process

Trade Transaction

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TradeFlow

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MessageBridge

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CreditCore

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Acknowledgement

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Control Platform

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└—— Validate acknowledgement

└—— Check duplicate

└—— Check processing time

└—— Compare amounts

└—— Compare utilization

└—— Reconcile status

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Exception?

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No Yes

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▼ ▼

Mark STP Create Case

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Calculate Priority

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Assign Queue

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Investigation

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Maker Resolution

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Critical Case?

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No Yes

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Close Checker Review

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Close

The major change is that exception management becomes **event-driven rather than reconciliation-driven**.

4. Stakeholder Map

Stakeholder	Primary Interest
Trade Operations	Fast exception investigation
Product Owner	Business value and prioritization
Credit Risk	Correct utilization
Compliance	Control evidence and auditability
Technology	Integration reliability
QA	Requirement and release quality
Production Support	Failure diagnosis
Management	Performance, risk and trends

RACI

Activity	Ops	PO	Risk	Compliance	Tech	QA	Support	Mgmt
Requirements	R	A	C	C	C	C	C	I
Business Rules	R	A	C	C	C	I	I	I
Solution Design	C	A	C	C	R	C	C	I
Development	I	C	I	I	A/R	C	C	I
Testing	C	C	C	I	C	A/R	C	I
UAT	A/R	C	C	C	C	C	C	I
Production Support	C	I	I	I	C	I	A/R	I
Management Reporting	C	C	C	C	I	I	I	A

5. Business Requirements Document

Objective

Create a centralized control mechanism for detecting and managing transaction-processing and credit-utilization exceptions between TradeFlow and CreditCore.

In Scope

- transaction monitoring;
- acknowledgement monitoring;
- duplicate detection;
- utilization reconciliation;
- exception creation;
- prioritization;
- assignment;
- SLA monitoring;
- maker-checker controls;
- notifications;
- audit history;
- operational reporting;
- management reporting.

Out of Scope

- payment processing;
- credit approval;
- facility creation;
- customer onboarding;
- sanctions screening;
- accounting;
- SWIFT message generation;
- automatic financial adjustment;
- autonomous exception closure.

Assumptions

- every transaction has a unique Transaction ID;

- Facility ID is available where credit utilization applies;
- timestamps are standardized;
- source systems provide required status events;
- authorized users can be mapped to roles;
- currency reference data is available.

Dependencies

- source-system feeds;
 - integration-layer availability;
 - reference data;
 - identity and access management;
 - notification service;
 - reporting infrastructure.
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Business Requirements

BR-01

The platform shall monitor transactions across the defined transaction lifecycle.

BR-02

The platform shall detect missing acknowledgements.

BR-03

The platform shall detect duplicate transaction messages.

BR-04

The platform shall identify transaction and utilization mismatches.

BR-05

The platform shall create an exception automatically when a defined rule fails.

BR-06

Exceptions shall be prioritized according to business impact.

BR-07

Exceptions shall be assigned to appropriate operational queues.

BR-08

The platform shall monitor SLA deadlines.

BR-09

Critical exceptions shall require maker-checker approval before closure.

BR-10

Users shall be able to view the complete transaction history.

BR-11

All material user actions shall be auditable.

BR-12

Management shall have access to exception and reconciliation reporting.

BR-13

The platform shall support daily reconciliation.

BR-14

Access shall be controlled by user role.

BR-15

Automated exception summaries shall remain advisory and require human review.

6. Functional Specification

Screen 1 – Operations Dashboard

Displays:

- transactions processed today;
 - STP rate;
 - open exceptions;
 - critical exceptions;
 - SLA-at-risk cases;
 - breached cases;
 - reconciliation status;
 - exceptions by category.
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Screen 2 – Exception Work Queue

Fields:

- Exception ID

- Transaction ID
- Facility ID
- Exception Type
- Amount
- Currency
- Priority
- Created Time
- SLA Due Time
- Assigned Team
- Assigned User
- Status

Filters:

- priority;
- exception type;
- age;
- currency;
- amount;
- status;
- assigned user;
- SLA status.

Screen 3 – Transaction Details

Displays the complete transaction journey:

TradeFlow

10:01:03 - Transaction Created

MessageBridge

10:01:07 - Message Published

CreditCore

10:01:12 - Message Received

CreditCore

10:01:14 - Utilization Updated

Acknowledgement

10:01:16 - Success

If an expected event is absent, it is highlighted.

Screen 4 – Reconciliation

Shows:

Transaction	Trade Amount	Credit Utilization	Difference	Status
TX1001	500,000	500,000	0	Matched
TX1002	750,000	700,000	50,000	Exception

Screen 5 – SLA Dashboard

Displays:

- cases within SLA;
- cases approaching SLA;
- breached cases;
- average resolution time;
- resolution time by exception type;
- resolution time by team.

Screen 6 – Management Report

Displays trends rather than individual operational cases.

Management can see:

- STP trend;
- exception trend;
- top root causes;
- SLA trend;
- reconciliation accuracy;
- repeat incidents;
- exceptions by source.

7. Business Rules Catalogue

Rule	Definition
BRULE-01	Flag a transaction when acknowledgement is not received within 5 minutes.
BRULE-02	Reject an exact duplicate containing the same Transaction ID and message identifier.
BRULE-03	Create an exception when transaction amount and utilization amount differ beyond the configured tolerance.
BRULE-04	High-value discrepancies must be escalated within 15 minutes.
BRULE-05	Critical exceptions require maker-checker approval.
BRULE-06	Maker and checker cannot be the same user.
BRULE-07	Closed exceptions cannot be edited without reopening them.
BRULE-08	Reopened cases must retain previous investigation history.

Rule	Definition
BRULE-09	SLA timers begin when an exception is created.
BRULE-10	System downtime periods approved as maintenance may be excluded from applicable SLA calculations.
BRULE-11	Every manual status change must capture user, timestamp and reason.
BRULE-12	Automated summaries cannot change transaction status or close an exception.

8. Data Dictionary

Field	Type	Description
Transaction_ID	VARCHAR	Unique transaction identifier
Facility_ID	VARCHAR	Associated lending facility
Message_ID	VARCHAR	Unique integration message
Transaction_Amount	DECIMAL	Trade transaction amount
Currency	VARCHAR(3)	ISO currency
Utilization_Amount	DECIMAL	Amount recorded against facility
Created_Timestamp	DATETIME	Transaction creation time
Sent_Timestamp	DATETIME	Message transmission time
Received_Timestamp	DATETIME	Downstream receipt
Ack_Timestamp	DATETIME	Acknowledgement time
Message_Status	VARCHAR	Current processing status
Exception_ID	VARCHAR	Exception identifier
Exception_Type	VARCHAR	Classification
Priority	VARCHAR	Critical/High/Medium/Low

Field	Type	Description
Assigned_To	VARCHAR	Current case owner
SLA_Due	DATETIME	Resolution deadline
Exception_Status	VARCHAR	Case lifecycle status

9. Source-to-Target Mapping

Source	Source Field	Target	Target Field	Transformation
TradeFlow	Trade_ID	Control Platform	Transaction_ID	Direct
TradeFlow	Trade_Value	Control Platform	Transaction_Amount	Decimal normalization
TradeFlow	CCY	Control Platform	Currency	ISO validation
MessageBridge	Msg_ID	Control Platform	Message_ID	Direct
MessageBridge	Msg_Time	Control Platform	Sent_Timestamp	UTC normalization
CreditCore	Facility_No	Control Platform	Facility_ID	Direct
CreditCore	Used_Amount	Control Platform	Utilization_Amount	Decimal normalization
CreditCore	Update_Time	Control Platform	Received_Timestamp	UTC normalization

10. User Stories & Acceptance Criteria

US-01 – Monitor Transaction Journey

As an Operations Analyst, I want to see the complete transaction journey so that I can identify where processing stopped.

Acceptance Criteria

Given a valid Transaction ID
When I open Transaction Details
Then I can see every available processing event in chronological order.

US-02 – Missing Acknowledgement

As an Operations Analyst, I want an exception created when an acknowledgement is missing so that delayed transactions are investigated quickly.

Acceptance Criteria

Given a transaction message has been sent
When no acknowledgement is received within five minutes
Then a Missing Acknowledgement exception is created automatically.

US-03 – Duplicate Detection

As an Operations Analyst, I want duplicate messages detected so that the same transaction is not processed twice.

US-04 – Utilization Mismatch

As a Credit Risk user, I want transaction and utilization values compared so that credit discrepancies are visible.

US-05 – Exception Priority

As an Operations Manager, I want exceptions prioritized so that the most material issues are investigated first.

US-06 – Work Assignment

As an Operations Manager, I want exceptions assigned to queues and users so that ownership is clear.

US-07 – SLA Warning

As an Operations Analyst, I want advance warning when a case approaches its SLA so that I can act before it breaches.

US-08 – Escalation

As an Operations Manager, I want breached critical cases escalated automatically so that unresolved risks receive attention.

US-09 – Maker-Checker

As a Control user, I want critical exceptions independently reviewed so that one person cannot resolve and approve the same critical case.

US-10 – Reconciliation

As an Operations Analyst, I want daily transactions reconciled so that missing and mismatched records can be identified.

US-11 – Audit Trail

As a Compliance user, I want a complete audit trail so that I can determine who performed each action and when.

US-12 – Management Reporting

As a Manager, I want exception trends and root causes so that recurring operational problems can be addressed.

US-13 – Search

As a Support Analyst, I want to search by Transaction ID, Facility ID and Message ID so that investigations can begin from whichever identifier is available.

US-14 – Role-Based Access

As a System Administrator, I want permissions based on user roles so that users only perform authorized actions.

US-15 – Exception Summary

As an Operations Analyst, I want a concise summary of a complicated exception so that I can understand the investigation history faster.

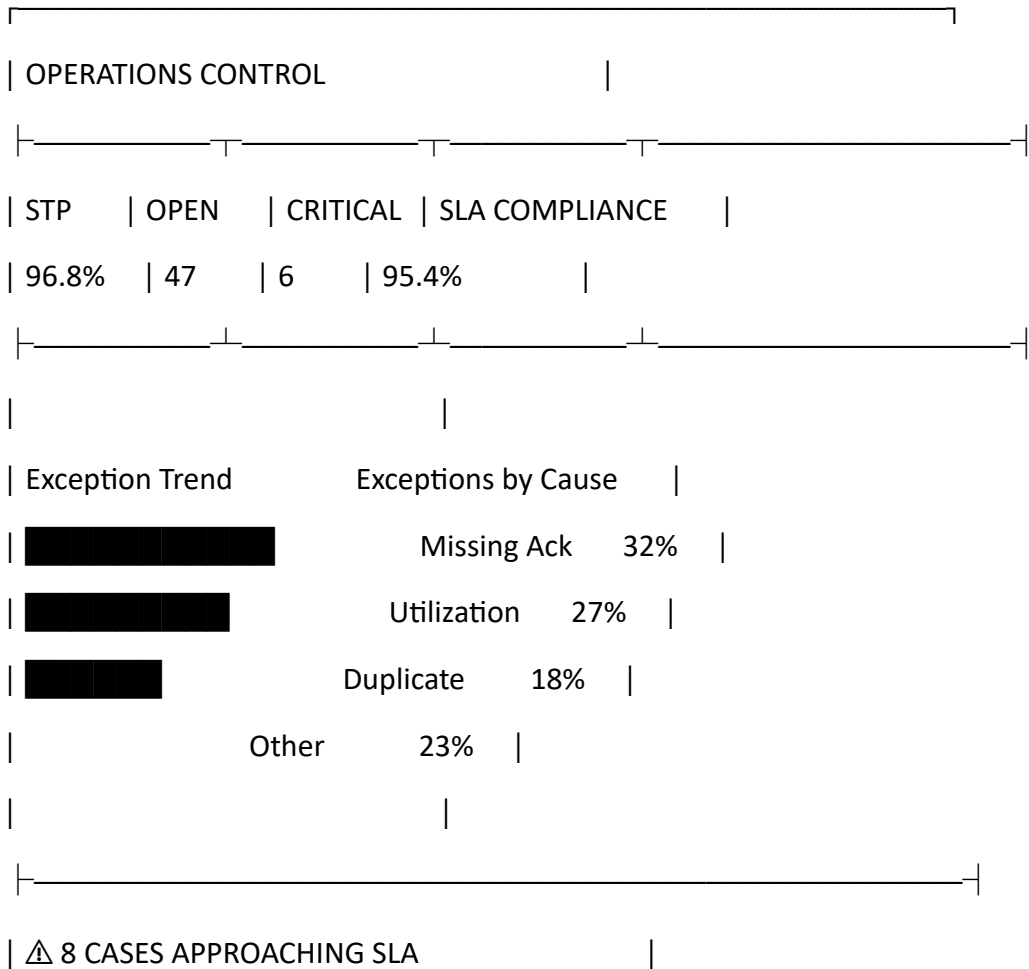
Acceptance Criteria

The summary must:

- identify itself as system-generated;
- reference available case information only;
- remain editable/reviewable by the investigator;
- not approve financial adjustments;
- not close cases;
- not override maker-checker controls.

11. Prototype

Operations Dashboard



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Exception Queue

EXCEPTION WORK QUEUE						

ID	TXN	TYPE	PRIORITY	AGE	STATUS	

EX001	TX1055	Utilization	Critical	12m	Assigned	
EX002	TX1062	Missing Ack	High	8m	New	
EX003	TX1068	Duplicate	Medium	6m	Review	

12. Requirements Traceability Matrix

BR	User Story	Rule	Test Case Status	
BR-01	US-01	—	TC-01	Passed
BR-02	US-02	BRULE-01	TC-02	Passed
BR-03	US-03	BRULE-02	TC-03	Passed
BR-04	US-04	BRULE-03	TC-04	Passed
BR-05	US-02/03/04	Multiple	TC-05	Passed
BR-06	US-05	BRULE-04	TC-06	Passed
BR-07	US-06	—	TC-07	Passed
BR-08	US-07/08	BRULE-09	TC-08	Passed
BR-09	US-09	BRULE-05/06	TC-09	Passed
BR-10	US-01	—	TC-10	Passed
BR-11	US-11	BRULE-11	TC-11	Passed
BR-12	US-12	—	TC-12	Passed

BR	User Story	Rule	Test Case Status	
BR-13	US-10	—	TC-13	Passed
BR-14	US-14	—	TC-14	Passed
BR-15	US-15	BRULE-12	TC-15	Passed

Statuses above represent simulated portfolio UAT results, not production testing.

13. UAT Package

Severity Definitions

Critical

Control failure, material financial exposure or inability to process critical transactions.

High

Major function unavailable with significant operational impact.

Medium

Function works partially or has a reasonable workaround.

Low

Minor usability or presentation issue.

Sample Test Cases

TC-02 – Missing Acknowledgement

Precondition: Valid transaction transmitted.

Action: Do not generate downstream acknowledgement.

Expected Result: Exception automatically created after five minutes and routed according to priority rules.

TC-03 – Duplicate

Send the same Transaction ID and Message ID twice.

Expected Result: Second message is identified as an exact duplicate and prevented from normal downstream processing.

TC-04 – Utilization Mismatch

Trade Amount: 1,000,000

Utilization Amount: 950,000

Expected Result: 50,000 discrepancy identified and exception created.

Sample Defect

DEF-007

Title: SLA timer resets when exception is reassigned.

Severity: High

Expected: Original SLA deadline remains unchanged.

Actual: Reassignment creates a new SLA deadline.

Decision: Release blocker because reassignment could hide genuine SLA breaches.

Sign-Off Criteria

- 100% Critical tests passed;
 - 100% High-priority requirements tested;
 - no open Critical defects;
 - no open High defects without approved workaround;
 - reconciliation results validated;
 - Risk and Operations approval obtained;
 - regression testing completed.
-

14. SQL Validation Pack

Find Exact Duplicates

SELECT

transaction_id,

message_id,

COUNT(*) AS occurrence_count

```
FROM transaction_messages  
GROUP BY transaction_id, message_id  
HAVING COUNT(*) > 1;
```

Find Transactions Missing Downstream Records

```
SELECT  
    t.transaction_id,  
    t.transaction_amount,  
    t.currency  
FROM trade_transactions t  
LEFT JOIN credit_utilization c  
    ON t.transaction_id = c.transaction_id  
WHERE c.transaction_id IS NULL;
```

Find Utilization Mismatches

```
SELECT  
    t.transaction_id,  
    t.transaction_amount,  
    c.utilization_amount,  
    t.transaction_amount - c.utilization_amount AS difference  
FROM trade_transactions t  
JOIN credit_utilization c  
    ON t.transaction_id = c.transaction_id  
WHERE ABS(t.transaction_amount - c.utilization_amount) > 0.01;
```

Find Late Acknowledgements

```
SELECT  
    transaction_id,  
    sent_timestamp,  
    ack_timestamp,  
    TIMESTAMPDIFF(MINUTE, sent_timestamp, ack_timestamp) AS delay_minutes
```

```
FROM transaction_messages
WHERE ack_timestamp IS NULL
      OR TIMESTAMPDIFF(MINUTE, sent_timestamp, ack_timestamp) > 5;
```

Daily Reconciliation

```
SELECT
    DATE(t.created_timestamp) AS business_date,
    COUNT(DISTINCT t.transaction_id) AS trade_count,
    COUNT(DISTINCT c.transaction_id) AS credit_count,
    COUNT(DISTINCT t.transaction_id)
      - COUNT(DISTINCT c.transaction_id) AS count_difference
FROM trade_transactions t
LEFT JOIN credit_utilization c
    ON t.transaction_id = c.transaction_id
GROUP BY DATE(t.created_timestamp);
```

15. Power BI Dashboard

The management dashboard would track six primary measures.

Straight-Through Processing Rate

STP Rate = Transactions completed without manual intervention / Total eligible transactions

Exception Rate

Exception Rate = Transactions creating exceptions / Total transactions

Average Resolution Time

Average time between exception creation and final resolution.

SLA Compliance

Percentage of applicable exceptions resolved within their agreed SLA.

Reconciliation Accuracy

Percentage of transactions successfully matched between TradeFlow and CreditCore.

Exceptions by System and Cause

Used to distinguish isolated operational cases from recurring system problems.

The dashboard should also show trends.

A 3% exception rate by itself provides limited information.

A movement from **1.1% → 1.5% → 2.2% → 3.0%** is much more useful because it tells management that something is changing.

16. Change Request & Impact Analysis

CR-001 – New Message Format

Request

Technology proposes replacing the existing message structure with a new version containing additional facility and processing attributes.

Reason

The downstream platform requires additional information to support enhanced utilization processing.

Impact Assessment

Requirements

Message validation requirements need updating.

Integration

MessageBridge mappings must support the new structure.

Data

New attributes need to be added to the source-to-target mapping.

Business Rules

Duplicate detection may need to consider the new message identifier.

Testing

Integration, regression, reconciliation and negative test cases must be updated.

Reporting

Any newly reportable fields require semantic-model assessment.

Support

Runbooks and troubleshooting guidance require updates.

Recommendation

Support both formats during a controlled transition period rather than performing an immediate cutover.

This reduces migration risk and allows reconciliation between old and new processing behaviour.

17. Decisions and Trade-Offs

What I Deprioritized

Predictive Exception Modelling

I would not include predictive modelling in the first release.

The first priority should be reliable deterministic controls.

If the platform cannot consistently identify a missing acknowledgement or utilization mismatch, adding a prediction model does not solve the core problem.

Decision: Later phase.

Automated Financial Correction

Operations requested faster resolution, but automatically adjusting financial values introduces considerably greater control risk.

Decision: Out of scope.

The platform identifies the discrepancy. An authorized human remains responsible for the financial action.

Mobile Application

Useful, but not necessary for the first release.

Decision: Deprioritized in favour of core monitoring and reconciliation.

18. Resolving a Stakeholder Conflict

A realistic conflict arose around exception thresholds.

Operations wanted broader tolerances to reduce low-value exception volumes.

Credit Risk wanted every discrepancy surfaced.

Both requests made sense from their respective perspectives.

I would not solve this by simply choosing one stakeholder's requirement.

Instead, I would separate **detection from escalation**.

Every discrepancy can be recorded for reconciliation and audit purposes.

The amount, materiality and risk rules then determine whether it becomes:

- informational;
- low priority;
- high priority;
- critical.

That preserves Risk's visibility without forcing Operations to treat every discrepancy as equally urgent.

19. Where Automation Stops

Automation is useful for:

- summarizing case history;
- organizing investigation notes;
- identifying similar historical exception descriptions;
- presenting likely areas for investigation;
- helping support teams navigate large amounts of case information.

It should not independently:

- approve credit;
- change facility utilization;
- release transactions;
- override control rules;
- approve material financial adjustments;
- close critical exceptions;
- act as the checker in maker-checker workflows.

For these actions, accountability matters more than convenience.

Human approval remains mandatory.

20. Measuring Success After Release

A successful deployment is not one where the software simply goes live.

I would review performance at approximately 30, 60 and 90 days.

The questions would be:

Are exceptions being detected earlier?

Has manual reconciliation effort decreased?

Is STP improving?

Are SLA breaches falling?

Are repeat exceptions reducing?

Is reconciliation accuracy improving?

Are users resolving cases faster?

Are the rules generating too many false positives?

I would also compare the projected benefits against actual results.

If the project predicted a 30% reduction in manual reconciliation but achieved 12%, the correct response is not to change the presentation.

It is to investigate the gap.

21. Expected Outcome

If implemented successfully, the proposed platform is expected to provide:

- earlier identification of transaction failures;
- better visibility into credit-utilization mismatches;
- clearer ownership of operational exceptions;
- stronger SLA monitoring;
- less manual reconciliation;
- better auditability;
- improved management visibility;

- better information for root-cause analysis.

The target is a **30% reduction in manual reconciliation effort**, but this remains a projected benefit until measured after implementation.

22. Five-Minute Demonstration Structure

0:00–0:45 — Problem

Explain how a Trade Finance transaction moves between systems and what happens when the transaction and utilization records stop matching.

0:45–1:20 — AS-IS

Show the current process and explain where manual investigation occurs.

1:20–2:00 — TO-BE

Show how centralized monitoring detects and creates exceptions.

2:00–3:00 — Prototype

Demonstrate:

Operations Dashboard → Exception Queue → Transaction Details → Reconciliation.

3:00–3:40 — Requirements & Controls

Show BRD, business rules, maker-checker control and RTM.

3:40–4:20 — Validation

Show SQL reconciliation and UAT examples.

4:20–4:40 — Reporting

Show STP, exception rate, SLA compliance and reconciliation accuracy.

4:40–5:00 — Outcome

Close with the projected business impact and explain how results would be measured after release.

My Role and Decisions

I approached this case study as a Technical Business Analyst rather than starting with a dashboard or a list of features.

My first question was:

Where does the business lose control of the transaction?

That led to the AS-IS analysis.

From there, I worked through the exception lifecycle, business rules, data requirements, user responsibilities, controls, testing and reporting.

Several decisions were deliberate:

- centralized monitoring instead of separate application-level views;
- deterministic controls before predictive models;
- risk-based prioritization rather than treating every exception equally;
- detection separated from escalation;
- maker-checker approval for critical cases;
- traceability from requirement through testing;
- human approval for financial and control decisions;
- measurable post-release targets rather than unsupported claims of impact.

The technical pieces matter, but the main purpose of the project is to show how I approach a complicated banking problem:

understand the process → identify the control gap → define the rules → design the workflow → validate the data → test the requirements → measure the result.

Portfolio Summary

Problem

Transaction failures between Trade Finance and lending systems can create delayed processing, duplicate messages, utilization mismatches and manual reconciliation.

Discovery

Mapped the transaction journey and identified fragmented monitoring, reactive exception detection and weak end-to-end visibility.

Analysis

Defined exception scenarios, stakeholders, business rules, data mappings, control requirements and operational priorities.

Decisions

Selected centralized exception management, deterministic controls, risk-based prioritization and maker-checker approval.

Requirements

Created 15 business requirements, 15 user stories, business rules, functional requirements and data mappings.

Prototype

Designed Operations, Exception Queue, Transaction Details, Reconciliation, SLA and Management views.

Validation

Created an RTM, UAT package and SQL reconciliation checks.

Measurable Outcome

Targeting:

≥95% STP

≥95% SLA compliance

≥99.5% reconciliation accuracy

30% projected reduction in manual reconciliation effort

Actual results would only be reported after implementation and post-release measurement.

Skills Demonstrated

Business Analysis: Requirements Elicitation, BRD, Functional Specification, User Stories, Acceptance Criteria, AS-IS/TO-BE, Gap Analysis, RACI, Business Rules, RTM

Banking: Trade Finance, Credit Utilization, Transaction Processing, Reconciliation, Operational Controls

Technical: SQL, Messaging Concepts, Data Mapping, Integration Analysis, Exception Management

Testing: UAT, Test Cases, Defect Management, Regression Impact Analysis

Reporting: Power BI, KPI Design, Operational Reporting

Delivery: Change Management, Release Impact Analysis, Stakeholder Management, Risk Assessment

Final Note

.

It demonstrates how I would analyze and design a Trade Finance exception-control solution based on my professional experience with banking processes, requirements, integrations, reconciliation, SQL validation, UAT and production-support environments.

It does not represent a Bank of America product or disclose any confidential employer information.

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